PROVEIT / RESEARCH CONTINUATION

Higher Digamma Powers and Ordered Gamma Sums

Exact finite parts, spectral zeta coefficients, and polylogarithmic depth cancellations

Prepared with ChatGPT for Vladimir Reshetnikov and the ProveIt project
11 October 2026

Abstract

We develop a complete finite-part formula for every positive integer power of the digamma function. A normalized Mittag–Leffler expansion, finite summation by parts, and a large-circle residue calculation reduce the unit-interval moment to an absolutely convergent harmonic sum and the linear Laurent coefficient of an explicit harmonic Dirichlet series. The entire boundary polynomial is evaluated; its constant is zero for odd powers and $(-1)^m\pi^{2m}/(2m+1)$ for power $2m$. A second form uses only an ordinary convergent series and an explicit finite list of ordinary Stieltjes constants. We carry out the fifth power completely, including the mixed double-zeta derivatives that do not occur at powers two through four, and give normalized primitives and differentiated polygamma identities for all powers.

A complementary two-marker Gamma formula sums ordered relative harmonic words over depth while retaining an arbitrary complex tagged exponent. Integer marker differences give finite Hurwitz-zeta reductions, and spectral differentiation produces generalized Stieltjes and log-Gamma identities. We also prove finite-interval formulas with several tagged exponents and polylogarithmic depth cancellations. Two precisely located external-source errors are corrected by direct proofs. Exact algebraic checks and independent high-precision diagnostics accompany the proofs. The Gaussian S_6 and revised S_8 conjectures remain unresolved.

Source baseline: ProveIt revision `b7667ae40a2957eed9a53d8559d53a3ee20dd1c1`.
The results are proposed additions to the inspected material; historical priority beyond the cited comparison has not been established.

Contents

1 Research targets, scope, and notation **1**
1.1 Conventions . . . 1

2 Pole coefficients and the normalized global expansion **2**
2.1 A finite recurrence for the coefficient differences . . . 3

3 The universal contour polynomial **4**

4 A finite-part formula for every positive integer power **5**
4.1 A closed ordinary-series form . . . 7

5 A finite algorithm in strict harmonic words **8**
5.1 Recovery of the lower-power formulas . . . 9

6 The fifth power with every finite correction **10**
6.1 An ordinary convergent formula for the same finite part . . . 11
6.2 Explicit kernels for the height-one coefficients . . . 12
6.3 The mixed derivative and its polylogarithmic representations . . . 13
6.4 What scalar polynomial cancellation proves . . . 13

7 Normalized primitives and polygamma identities **14**

8 Ordered depth generators and exact cancellations **15**
8.1 Relative ordered harmonic sums . . . 16
8.2 A two-marker theorem . . . 16
8.3 Integer gaps and ordinary zeta derivatives . . . 18
8.4 A finite-interval formula with several tagged exponents . . . 19
8.5 A polylogarithmic companion . . . 20

9 Source audit and precise corrections **21**
9.1 Scope of the repository audit . . . 21
9.2 A reversed pair of integrals in a Tornheim source . . . 21
9.3 A distinction between multiple and ordinary zeta reductions . . . 22
9.4 Unchanged conjecture status . . . 23

10 Verification and reproducibility **23**
10.1 Exact finite algebra . . . 23
10.2 An independent fifth-power calculation . . . 23
10.3 How to reproduce the package . . . 24

11 Further exact research questions **24**
11.1 The first mixed spectral coordinate . . . 24

11.2 Scalar polynomials and the growth of spectral support 25
11.3 Explicit sixth and seventh moments 25
11.4 A common Hurwitz shift with correct endpoint terms 25
11.5 Differential polynomials in digamma and polygamma 26
11.6 Several tags at independent endpoints 26
11.7 Boundary markers and removable Gamma singularities 26
11.8 Polylogarithmic cancellation at roots of unity 26
11.9 Gaussian relations beyond the searched family 26
11.10Formal verification of the analytic normalization 27

A A bounded investigation of the current Gaussian S_6 target **27**
A.1 A useful local attribution 28

References **28**

1 Research targets, scope, and notation

The immediate starting point is the quartic digamma calculation in the incoming reports [2, 3]. The first higher-power question in [2] asks for the fifth moment with *every* finite correction, including the correct linear Laurent coefficients and ordinary zeta derivatives. Its next question asks whether scalar polynomials in the digamma function can cancel lower-depth data at arbitrary degree. The present article resolves the first question and proves an all-power reduction mechanism. It also identifies the precise extra coordinate that must be addressed before the stronger scalar cancellation proposal can be asserted at degree five.

The second line of work begins with the relative ordered harmonic interpolation in the incoming independent-order report [4]. We prove a two-marker generating formula and exact depth cancellations that keep a tagged complex exponent, together with all its spectral derivatives. This provides a useful connection between the harmonic and polylogarithmic parts of the research program.

Source target	Result established here	Scope retained
Fifth digamma finite part	Complete formula in Theorem 6.1, with all scalar and spectral corrections	No reduction of the retained spectral constants to a smaller arithmetic algebra
Arbitrary integer powers	Normalized expansion, universal contour constant, finite word algorithm, and ordinary subtracted sum	Fixed integer power; no assertion for arbitrary complex powers
Canonical scalar polynomials	Explicit fifth-degree cancellation of height-one lower data; mixed derivative term remains	A formal coordinate statement, not arithmetic independence
Independent ordered exponents	Two-marker Gamma formula, integer-gap reductions, Stieltjes and Gamma consequences	Stated convergence discs; several tags are proved here on finite intervals
Gaussian S_6, revised S_8	A bounded expanded S_6 relation search is recorded separately	Both remain conjectural

The source review is targeted. It is not a proof audit of every chapter or every incoming archive. The familiar Gamma recurrences and asymptotics, finite stuffle algebra, and low-weight multiple-zeta evaluations are antecedents, not new results [5, 6]. Nor is the existence of entire harmonic interpolation new: we use the established interpolant of Ihara–Nakamura–Yamamoto [7]. The new contribution relative to the inspected repository is the collection of exact extensions, normalizations, and reductions proved below. No result is accepted on the basis of integer-relation recognition.

1.1 Conventions

Let $\psi(z) = \Gamma'(z)/\Gamma(z)$, let γ be Euler's constant, and put

$$H_n^{(r)} = \sum_{j=1}^{n} j^{-r}, \qquad H_n = H_n^{(1)}, \qquad H_n(w) = \sum_{n \geq j_1 > \cdots > j_d \geq 1} j_1^{-w_1} \cdots j_d^{-w_d}.$$

The empty word has value $H_n(\varnothing) = 1$; a nonempty word has $H_0(w) = 0$. All finite words used in the digamma calculation have positive integer entries. We use the strict multiple-zeta convention

$$Z(s_1, \ldots, s_d) = \zeta(s_1, \ldots, s_d) = \sum_{n_1 > \cdots > n_d \geq 1} n_1^{-s_1} \cdots n_d^{-s_d}. \tag{1.1}$$

In particular $Z(s,w)=\sum_{n\ge1}H_{n-1}(w)n^{-s}$ where the series converges. At the positive integer words below, a first entry at least two makes the series absolutely convergent.

Our Stieltjes convention is

$$\zeta(1+u,a)=\frac{1}{u}+\sum_{j\ge0}\frac{(-1)^j}{j!}\gamma_j(a)u^j,\qquad a>0,\qquad \gamma_j=\gamma_j(1). \tag{1.2}$$

Thus $\gamma_0(a)=-\psi(a)$ and $\gamma_0=\gamma$. For a meromorphic germ $f(u)$, $[u^j]f$ denotes its Laurent coefficient. In particular $[u]f$ is meaningful even at a pole and is not then $f'(0)$. This distinction is essential in the fifth-power formula.

Definition 1.1 (Original-coordinate finite part)**.** Suppose $f(x)=\sum_{j=1}^{k}a_jx^{-j}+g(x)$ near zero, with g continuous on $[0,1]$. Define

$$\mathrm{FP}\int_0^1 f(x)\,\mathrm{d}x=\lim_{\epsilon\downarrow0}\left\{\int_\epsilon^1 f(x)\,\mathrm{d}x-\sum_{j=2}^{k}\frac{a_j\epsilon^{1-j}}{j-1}+a_1\log\epsilon\right\}. \tag{1.3}$$

The coordinate is exactly x and the upper endpoint is exactly 1. We write $Q_k=\mathrm{FP}\int_0^1\psi(x)^k\,\mathrm{d}x$ for $k\ge1$.

For a sequence $B_N=P(\log N)+o(1)$, with P a polynomial, define $\mathrm{CT}_{\log N}B_N=P(0)$. This cutoff operation and a Laurent coefficient in an auxiliary variable are not identified by convention; Lemma 4.2 proves the comparison needed here. Complex logarithms in the global primitive use the principal branch on $\mathbb{C}\setminus(-\infty,0]$. All powers of positive real summation variables use the real logarithm.

2 Pole coefficients and the normalized global expansion

The standard Taylor expansion of $\psi(1+\epsilon)$ and the recurrence $\psi(z+1)=\psi(z)+1/z$ give [5, §§5.5,5.7]

$$\psi(-n+\epsilon)=-\frac{1}{\epsilon}+b_n+\sum_{r\ge2}c_{r,n}\epsilon^{r-1}, \tag{2.1}$$
$$b_n=H_n-\gamma,\qquad c_{r,n}=H_n^{(r)}+(-1)^r\zeta(r),\qquad n\ge0.$$

This is a convergent local expansion for $0<|\epsilon|<1$.

For each fixed positive integer k, define the finite polynomial data

$$p_{k,j}(n)=[\epsilon^{k-j}]\left(-1+b_n\epsilon+\sum_{r=2}^{k}c_{r,n}\epsilon^r\right)^k,\qquad 1\le j\le k, \tag{2.2}$$

$$r_k(n)=p_{k,1}(n),\qquad P_{k,n}(z)=\sum_{j=1}^{k}\frac{p_{k,j}(n)}{(z+n)^j},\qquad \kappa_k=[\epsilon^0]\psi(\epsilon)^k. \tag{2.3}$$

The symbol κ_k is a local Laurent constant; it will not be confused with the universal contour constant C_k introduced later. The pole coefficients satisfy

$$p_{k,j}(n)=O_k\big((1+\log(n+1))^{k-j}\big). \tag{2.4}$$

Indeed $b_n=O(1+\log(n+1))$ and every $c_{r,n}$ with $r\ge2$ is bounded. In a term of degree $k-j$ in ϵ, at most $k-j$ factors of b_n can occur.

Theorem 2.1 (Normalized Mittag–Leffler formula for every power)**.** *For $z \notin \{0, -1, -2, \dots\}$,*

$$\boxed{\psi(z)^k = P_{k,0}(z) + \kappa_k + \sum_{n\geq 1} (P_{k,n}(z) - P_{k,n}(0)).} \tag{2.5}$$

The series converges normally on compact sets avoiding the poles. In particular the entire remainder is exactly the displayed constant, with no uncomputed entire function.

Proof. Normal convergence of the candidate series follows from (2.4): on a fixed compact set, for sufficiently large n,

$$(z+n)^{-j} - n^{-j} = O(n^{-j-1}),$$

uniformly in z. It remains to determine the possible entire remainder. Put $f(w) = \psi(w)^k - P_{k,0}(w)$, so f is analytic at zero and $f(0) = \kappa_k$. Integrate

$$\frac{zf(w)}{w(w-z)}$$

around $|w| = R_N = N + \frac{1}{2}$, with positive orientation and N large.

We need a uniform estimate on the full circle, including near the negative real axis. On its right half, the usual sectorial digamma asymptotic gives $\psi(w) = O(1 + \log R_N)$. On its left half, use

$$\psi(w) = \psi(1-w) - \pi \cot(\pi w). \tag{2.6}$$

Here $1-w$ lies in the right half-plane and has size comparable to R_N. If $|\Im w| \geq 1$, the cotangent is uniformly bounded. If $|\Im w| < 1$, the points of this circle are a fixed positive distance from the integers for all sufficiently large N, because its real intersections are half-integers. The cotangent is again uniformly bounded. Thus $f(w) = O_k((1 + \log R_N)^k)$ on the whole contour, and the integral is $O_k((1 + \log R_N)^k / R_N) = o(1)$.

The residues at $w = z$ and $w = 0$ are $f(z)$ and $-f(0)$. At $w = -n$, the residue is $-P_{k,n}(z) + P_{k,n}(0)$, as follows by expanding $z/[w(w-z)] = 1/(w-z) - 1/w$ there. The residue theorem and passage to the limit give (2.5). This proves the normalization, not just agreement of the principal parts. □

2.1 A finite recurrence for the coefficient differences

Write $\Delta p_{k,j}(n) = p_{k,j}(n) - p_{k,j}(n-1)$ for $n \geq 1$, and $\Delta r_k(n) = \Delta p_{k,1}(n)$.

Proposition 2.2 (Exact difference recurrence)**.** *For $1 \leq j \leq k$,*

$$\Delta p_{k,j}(n) = \sum_{\ell=1}^{k-j} \binom{k}{\ell} \sum_{h=j}^{k-\ell} \binom{\ell + h - j - 1}{h - j} \frac{p_{k-\ell,h}(n-1)}{n^{\ell+h-j}}. \tag{2.7}$$

An empty sum is zero; in particular $\Delta p_{k,k}(n) = 0$.

Proof. The recurrence for ψ gives the local identity

$$\psi(-n+\epsilon) = \psi(-(n-1)+\epsilon) + \frac{1}{n-\epsilon}.$$

Expand its kth power by the binomial theorem. In the term with ℓ copies of $(n-\epsilon)^{-1}$, multiply the pole $p_{k-\ell,h}(n-1)\epsilon^{-h}$ by

$$(n-\epsilon)^{-\ell} = \sum_{q\geq 0} \binom{\ell+q-1}{q} \frac{\epsilon^q}{n^{\ell+q}}.$$

The coefficient of ϵ^{-j} has $q = h - j$, which gives (2.7). Only finitely many coefficients contribute. □

The recurrence proves, in particular, $\Delta p_{k,j}(n) = O_k((1+\log n)^k/n)$ for $j < k$. Sharper logarithmic powers are immediate, but are not needed for the exact identities. Formula (2.7) is also a convenient implementation specification: it involves only lower powers and exact rational binomial coefficients.

3 The universal contour polynomial

The subtraction constants in (2.5) are individually awkward harmonic expressions. Their combined cutoff asymptotic is much simpler than their separate expansions suggest.

Theorem 3.1 (All coefficients of the large-circle boundary term)**.** *For every fixed integer* $k \geq 1$,

$$\begin{aligned}\kappa_k - \sum_{n=1}^{N} P_{k,n}(0) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} (\log N + i\theta)^k \,\mathrm{d}\theta + o(1) \\ &= \sum_{m=0}^{\lfloor k/2 \rfloor} \binom{k}{2m} \frac{(-1)^m \pi^{2m}}{2m+1} (\log N)^{k-2m} + o(1). \end{aligned} \tag{3.1}$$

Consequently

$$C_k := \mathrm{CT}_{\log N}\left(\kappa_k - \sum_{n=1}^{N} P_{k,n}(0)\right) = \begin{cases} (-1)^m \pi^{2m}/(2m+1), & k = 2m, \\ 0, & k \text{ odd}. \end{cases} \tag{3.2}$$

Proof. The residue at zero of $\psi(w)^k/w$ is κ_k. At $-n$ it is $-P_{k,n}(0)$. Thus, on $R = N + \frac{1}{2}$, the residue theorem gives the exact identity

$$\kappa_k - \sum_{n=1}^{N} P_{k,n}(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \psi(Re^{i\theta})^k \,\mathrm{d}\theta. \tag{3.3}$$

We compare this with the angular average of the principal logarithm. On the right half-circle, $\psi(w) - \operatorname{Log} w = O(1/R)$ uniformly. On the left half-circle with $|\Im w| \geq \log R$, reflection (2.6) and the exponential expansion of cotangent give

$$\psi(w) - \operatorname{Log} w = O(1/R) + O(e^{-2\pi|\Im w|}).$$

For example, in the upper left quadrant, $\operatorname{Log}(1-w) = \operatorname{Log} w - i\pi + O(1/R)$ and $\pi\cot\pi w = -i\pi + O(e^{-2\pi\Im w})$; the lower quadrant is the conjugate branch calculation.

The remaining arc near the negative real axis has angular length $O(\log R/R)$. On that arc, the bounded-cotangent argument in Theorem 2.1 gives $\psi(w) - \operatorname{Log} w = O(1)$, while both terms are $O(1+\log R)$. Factoring the difference of the kth powers shows that its contribution to the angular average is $O_k((1+\log R)^k/R)$. The rest of the circle contributes a quantity of the same or smaller order. The logarithm has a jump at the one endpoint of the angular parametrization, which does not affect the integral. Therefore

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \left(\psi(Re^{i\theta})^k - (\log R + i\theta)^k\right) \mathrm{d}\theta = o(1).$$

Replacing $\log R$ by $\log N$ changes this by $o(1)$. Odd powers of θ integrate to zero and the even angular moments are $(-1)^m\pi^{2m}/(2m+1)$, proving both assertions. □

For orientation, $C_1 = C_3 = C_5 = 0$, $C_2 = -2\zeta(2)$, and $C_4 = 18\zeta(4)$. These are combined boundary constants. They are not the local Laurent constants κ_k.

4 A finite-part formula for every positive integer power

Define

$$J_k = \sum_{n\geq 1} \sum_{j=2}^{k} \frac{\Delta p_{k,j}(n)}{(j-1)n^{j-1}}, \tag{4.1}$$

$$D_k(s) = \sum_{n\geq 1} \frac{\Delta r_k(n)}{n^s}, \qquad \Re s > 0. \tag{4.2}$$

The first series is ordinary and absolutely convergent, by Proposition 2.2. The second converges absolutely in its initial half-plane. Its meromorphic continuation to a neighborhood of zero will follow from the proof below.

Theorem 4.1 (The all-power harmonic spectral formula)**.** *For every integer $k \geq 1$,*

$$\boxed{Q_k = C_k + J_k + [s]D_k(s).} \tag{4.3}$$

All data on the right are specified by the finite polynomial calculation (2.2), the exact difference recurrence (2.7), and the coefficient of a meromorphic germ. The identity uses the original-coordinate finite part (1.3).

We first record the exact cutoff comparison that determines the sign and the order of the spectral coefficient.

Lemma 4.2 (Logarithmic cutoffs and Laurent coefficients)**.** *Suppose $a_n = P(\log n)+O(n^{-\delta}(1+\log n)^M)$, where P is a polynomial and $\delta > 0$. Let $A(s) = \sum_{n\geq 1} a_n n^{-s}$ in its right half-plane of absolute convergence. Then $A(1+u)$ continues meromorphically near $u = 0$, and*

$$\mathrm{CT}_{\log N} \sum_{n=1}^{N} \frac{a_n}{n} = [u^0]A(1+u), \tag{4.4}$$

$$\mathrm{CT}_{\log N} \sum_{n=1}^{N} \frac{a_n \log n}{n} = -[u]A(1+u). \tag{4.5}$$

Proof. For $P(L) = \sum_j c_j L^j$, write

$$A(1+u) = \sum_j c_j(-1)^j \zeta^{(j)}(1+u) + \sum_{n\geq 1} \frac{a_n - P(\log n)}{n^{1+u}}.$$

The last series and all its derivatives converge normally for $\Re u > -\delta$. The elementary Stieltjes limit formula gives

$$\sum_{n=1}^{N} \frac{(\log n)^j}{n} = \frac{(\log N)^{j+1}}{j+1} + \gamma_j + O((1+\log N)^j/N).$$

The stated error follows by one sum–integral comparison applied to the derivative of $(\log x)^j/x$ on the tail. On the Laurent side, $[u^0](-1)^j\zeta^{(j)}(1+u) = \gamma_j$ and $[u](-1)^j\zeta^{(j)}(1+u) = -\gamma_{j+1}$. Termwise comparison proves the first formula and, after inserting one further factor of $\log n$, the second. Absolute convergence handles the subtracted remainder in both cases. □

Proof of Theorem 4.1. The harmonic-number asymptotic expansions, or reflection followed by the positive-axis digamma expansion, show that

$$r_k(n) = R_k(\log n) + O_k((1+\log n)^M/n), \tag{4.6}$$

$$\Delta r_k(n) = \frac{R_k'(\log n)}{n} + O_k((1+\log n)^M/n^2), \tag{4.7}$$

$$R_k(L) = \operatorname*{Res}_{\epsilon=0}(L - \pi\cot\pi\epsilon)^k. \tag{4.8}$$

Here and below M may depend on k. To justify the second estimate, use the complete harmonic asymptotic expansions through the $1/n$ term: $r_k(n) = R_k(\log n) + n^{-1}U_k(\log n) + O((1+\log n)^M/n^2)$ with a polynomial U_k. The difference of the explicit $n^{-1}U_k$ terms is $O((1+\log n)^M/n^2)$, and the leading difference is $R_k'(\log n)/n$ to that same accuracy. Thus the second formula does not follow from differencing an unspecified first-order error. It also follows directly from (2.7) and the first two harmonic asymptotic terms.

Apply Lemma 4.2 to $a_n = n\Delta r_k(n)$. It proves the meromorphic continuation of D_k and gives

$$\mathrm{CT}_{\log N} \sum_{n=1}^{N} \Delta r_k(n)\log n = -[s]D_k(s). \tag{4.9}$$

Now integrate the normally convergent series (2.5). After the pole $P_{k,0}$ is removed, the same $O(n^{-j-1})$ estimate proves uniform convergence also on the closed interval $[0,1]$, so integration across its regular endpoint is justified. The finite part of the pole at zero contributes $-\sum_{j=2}^{k} p_{k,j}(0)/(j-1)$; its simple pole has finite part zero. For a pole at $-n$, the integrated simple-pole term is $r_k(n)\log(1+1/n)$, and the higher-pole terms have numerators $p_{k,j}(n)(n^{1-j} - (n+1)^{1-j})/(j-1)$. The constant subtraction integrates to $-P_{k,n}(0)$.

Use the following *finite* summation identities:

$$\sum_{n=1}^{N} p_{k,j}(n)(n^{1-j} - (n+1)^{1-j}) = p_{k,j}(0) + \sum_{n=1}^{N} \frac{\Delta p_{k,j}(n)}{n^{j-1}} - \frac{p_{k,j}(N)}{(N+1)^{j-1}}, \qquad j \geq 2, \tag{4.10}$$

$$\sum_{n=1}^{N} r_k(n)\log(1+1/n) = r_k(N)\log(N+1) - \sum_{n=1}^{N} \Delta r_k(n)\log n. \tag{4.11}$$

The first terms in (4.10) cancel the finite part of the pole at zero exactly, while its final terms tend to zero. If $J_{k,N}$ denotes the cutoff in (4.1), we obtain

$$Q_k = \lim_{N\to\infty}\Bigg\{\kappa_k - \sum_{n=1}^{N} P_{k,n}(0) + J_{k,N} + r_k(N)\log(N+1) - \sum_{n=1}^{N} \Delta r_k(n)\log n\Bigg\}. \tag{4.12}$$

All terms here arose from finite sums or normally convergent series. By (4.6), the polynomial asymptotic of the boundary $r_k(N)\log(N+1)$ is $(\log N)R_k(\log N)$, which has no constant term. Theorem 3.1 supplies the constant C_k for the first two terms of (4.12), and (4.9) supplies $+[s]D_k$. Finally $J_{k,N} \to J_k$. Taking the constant coefficient of the complete cutoff expression proves (4.3). □

4.1 A closed ordinary-series form

The residue polynomial in (4.8) itself has a closed form. This makes every Stieltjes correction explicit, without naming any multiple-zeta Laurent coefficient.

Lemma 4.3 (Closed asymptotic residue polynomial)**.** *For an integer $k \geq 0$, define*

$$R_k(L) = \operatorname*{Res}_{t=0}(L - \pi\cot(\pi t))^k.$$

Then

$$\boxed{R_k(L) = -\frac{(L+i\pi)^k - (L-i\pi)^k}{2i\pi}.} \tag{4.13}$$

For $k \geq 1$, if $R_k'(L) = \sum_{j\geq 0} a_{k,j}L^j$, the ordinary Stieltjes contribution is the explicit finite expression

$$\boxed{-\sum_{j\geq 0} a_{k,j}\gamma_{j+1} = k\sum_{m=0}^{\lfloor (k-2)/2\rfloor}(-1)^m\binom{k-1}{2m+1}\pi^{2m}\gamma_{k-2m-1}.} \tag{4.14}$$

The right side is understood as zero for $k = 1$. Here $\zeta(1+s) = s^{-1} + \sum_{j\geq 0}(-1)^j\gamma_j s^j/j!$.

Proof. Put $f(t) = L - \pi\cot(\pi t)$. The elementary differential identity

$$f'(t) = (f(t) - L)^2 + \pi^2$$

and the zero residue of a meromorphic derivative give, for $k \geq 2$,

$$0 = \frac{1}{k-1}\operatorname*{Res}_{t=0}\frac{\mathrm{d}}{\mathrm{d}t}f(t)^{k-1} = R_k(L) - 2LR_{k-1}(L) + (L^2+\pi^2)R_{k-2}(L).$$

Since $R_0 = 0$ and $R_1 = -1$, solving this recurrence with characteristic roots $L \pm i\pi$ proves (4.13). Expanding its numerator yields

$$R_k(L) = -\sum_{m=0}^{\lfloor (k-1)/2\rfloor}(-1)^m\binom{k}{2m+1}\pi^{2m}L^{k-2m-1}.$$

On differentiating in L, the possible term with $k-2m-1 = 0$ vanishes. Use $(k-2m-1)\binom{k}{2m+1} = k\binom{k-1}{2m+1}$. This gives (4.14). □

To see directly why this finite expression is the Stieltjes term, suppose the residue difference satisfies the previously established asymptotic formula

$$r_k(n) - r_k(n-1) = \frac{R_k'(\log n)}{n} + O\left(\frac{(1+\log n)^M}{n^2}\right).$$

For $D_k(s) = \sum_{n\geq 1}(r_k(n) - r_k(n-1))n^{-s}$, subtraction therefore gives

$$D_k(s) = \sum_{j\geq 0} a_{k,j}(-1)^j\zeta^{(j)}(1+s) + \sum_{n\geq 1}\left(r_k(n) - r_k(n-1) - \frac{R_k'(\log n)}{n}\right)n^{-s}.$$

The second series and its first spectral derivative converge normally near $s = 0$. Since $[s](-1)^j\zeta^{(j)}(1+s) = -\gamma_{j+1}$, it follows that

$$\begin{aligned}[s]D_k(s) = k&\sum_{m=0}^{\lfloor (k-2)/2\rfloor}(-1)^m\binom{k-1}{2m+1}\pi^{2m}\gamma_{k-2m-1}\\ &-\sum_{n\geq 1}\left(r_k(n) - r_k(n-1) - \frac{R_k'(\log n)}{n}\right)\log n.\end{aligned} \tag{4.15}$$

The final sum is absolutely convergent. For example, the finite Stieltjes terms at powers four and five are respectively $12\gamma_3 - 24\zeta(2)\gamma_1$ and $20\gamma_4 - 120\zeta(2)\gamma_2$.

Corollary 4.4 (Ordinary subtracted series at every power)**.** *With the definitions above,*

$$Q_k = C_k + J_k + k \sum_{m=0}^{\lfloor (k-2)/2 \rfloor} (-1)^m \binom{k-1}{2m+1} \pi^{2m} \gamma_{k-2m-1} - \sum_{n \geq 1} \left(\Delta r_k(n) - \frac{R_k'(\log n)}{n} \right) \log n. \tag{4.16}$$

The finite Stieltjes sum is zero at $k = 1$. Both infinite sums are ordinary absolutely convergent sums. The last summand is $O_k((1 + \log n)^{M+1}/n^2)$.

This is useful both conceptually and computationally. The right side contains no unspecified regularization and no uncomputed entire remainder. Further terms of the same harmonic asymptotic expansions give convergent Hurwitz-zeta tail accelerations if numerical evaluation is wanted. Such acceleration is optional for the identity itself.

5 A finite algorithm in strict harmonic words

The spectral formula becomes finite algebra once its target class is specified. Work over $\mathbb{Q}[\gamma, \zeta(2), \ldots, \zeta(k)]$ and put $h_r = H_{n-1}^{(r)}$, $v = 1/n$. Compute the polynomial $p_{k,j}$ from (2.2), and replace $H_n^{(r)}$ by $h_r + v^r$ to obtain its value at n. Subtract its value at $n-1$. This produces every $\Delta p_{k,j}$ as a finite polynomial in $v, h_1, \ldots, h_k$.

The finite stuffle recursion, with (a, A) denoting a leading letter followed by a word, is

$$(a, A) * (b, B) = (a, A * (b, B)) + (b, (a, A) * B) + (a + b, A * B). \tag{5.1}$$

It expresses products of the h_r as a finite integer combination of strict words $H_{n-1}(w)$. The coefficients count equal-index possibilities as well as the two strict orderings. Thus

$$h_1^2 = 2H_{n-1}(1, 1) + H_{n-1}^{(2)},$$

and a further multiplication gives

$$h_1^3 = 6H_{n-1}(1, 1, 1) + 3H_{n-1}(2, 1) + 3H_{n-1}(1, 2) + H_{n-1}^{(3)}.$$

These are finite identities before any infinite sum is taken.

Every positive-integer word has a polynomial harmonic asymptotic

$$H_{n-1}(w) = B_w(\log n) + O((1 + \log n)^M/n) \tag{5.2}$$

for some polynomial B_w and integer M. Here is a direct induction that also supplies the continuation needed term by term. For the empty word the assertion is immediate. Given it for w, use $H_N(a, w) = \sum_{m=1}^{N} H_{m-1}(w) m^{-a}$. If $a = 1$, the Stieltjes cutoff formula applied to each power of $\log m$ and the absolutely convergent subtracted remainder give a polynomial in $\log N$ with an $O((1 + \log N)^{M'}/N)$ error. If $a \geq 2$, the sum converges and its tail has that same error form, so the polynomial is a constant. Replacing N by $n - 1$ preserves the claimed form. Consequently each $Z(1 + s, w)$ has a meromorphic germ at zero by Lemma 4.2; for first entry at least two its defining series is already holomorphic near zero.

Theorem 5.1 (Terminating harmonic-word reduction)**.** *At a fixed positive integer power k, the following procedure terminates and yields the exact formula (4.3):*

1. *Compute $p_{k,j}$ and their finite differences by (2.2) or (2.7).*
2. *Expand the summand of J_k and the polynomial $\Delta r_k(n)$ in $v, h_1, \dots, h_k$.*
3. *Apply the finite stuffle recursion to every harmonic monomial.*
4. *Replace $v^q H_{n-1}(w)$ by $Z(q, w)$ for J_k, and by $Z(q+s, w)$ for $D_k(s)$.*

Every first entry in J_k is at least two. Every first entry in $D_k(0)$ is at least one. The resulting word weights are at most $k-1$; coefficient factors in γ and ordinary zeta values account for the remaining formal weight. The prescription introduces only a finite number of specified linear Laurent coefficients.

Proof. Coefficient extraction and finite differences are finite polynomial operations. Each application of the stuffle recursion reduces the combined remaining word length, so it terminates. By (2.7), every monomial in $\Delta p_{k,j}$ contains at least one v, and division by n^{j-1} gives $q \geq j \geq 2$ in J_k. For Δr_k, it gives $q \geq 1$. Therefore all replacements in the initial half-planes are identities of absolutely convergent series. Their continuations at zero agree by the meromorphic subtraction justified by (5.2). Finally assign weight 1 to h_1, γ, v, weight r to $h_r, \zeta(r)$, and use (2.2). Both relevant summand polynomials have weight $k-1$, and stuffle preserves it. □

This is a reduction to a precise finite class of multiple-zeta values and their spectral coefficients. It does not prove that all those values reduce further to products of ordinary zeta values. For higher powers it is appropriate to retain admissible multiple-zeta words unless an exact additional relation is supplied.

5.1 Recovery of the lower-power formulas

Define the height-one coefficients

$$\eta = [u]Z(1+u, 1), \qquad \theta = [u]Z(1+u, 1, 1), \qquad \chi = [u]Z(1+u, 1, 1, 1). \tag{5.3}$$

The first instances of Theorem 4.1 are

$$\begin{aligned} Q_1 &= 0, \\ Q_2 &= 2\gamma_1 - 2\zeta(2), && (5.4)\\ Q_3 &= 3\zeta(2) - 6\eta - 6\gamma\gamma_1, && (5.5)\\ Q_4 &= 12\zeta(3) - 12\gamma\zeta(2) + 18\zeta(4) + 4\zeta'(3) - 24\theta + 24\gamma\eta + 12(\gamma^2 - \zeta(2))\gamma_1. && (5.6) \end{aligned}$$

The cubic and quartic identities are consistency checks against the incoming reports [2, 3]; they are not presented as new evaluations here. For example,

$$\begin{aligned} D_2(s) &= -2\zeta(1+s), & J_2 &= 0, \\ D_3(s) &= -6Z(1+s, 1) + 6\gamma\zeta(1+s), & J_3 &= 3\zeta(2), \\ D_4(s) &= -24Z(1+s, 1, 1) + 24\gamma Z(1+s, 1) \\ &\quad + 12(\zeta(2) - \gamma^2)\zeta(1+s) + 4\zeta(3+s), & J_4 &= 12\zeta(3) - 12\gamma\zeta(2). \end{aligned}$$

At $k = 1$ the residue is the constant -1, hence $J_1 = D_1 = 0$; the finite part also follows directly from $\log\Gamma(1) = 0$ and $\log\Gamma(\epsilon) = -\log\epsilon + O(\epsilon)$.

6 The fifth power with every finite correction

For compactness write $\zeta_r = \zeta(r)$ in this section. At a fixed pole use $b = b_n$, $c = c_{2,n}$, $d = c_{3,n}$, and $e = c_{4,n}$. Direct coefficient extraction gives

$$\begin{aligned} p_{5,5} &= -1, \qquad p_{5,4} = 5b, \qquad p_{5,3} = 5c - 10b^2, \\ p_{5,2} &= 10b^3 - 20bc + 5d, \\ r_5 &= -5b^4 + 30b^2c - 10c^2 - 20bd + 5e. \end{aligned} \tag{6.1}$$

For reference, the local constant is

$$\kappa_5 = -\gamma^5 + 20\gamma^3\zeta_2 - 55\gamma\zeta_4 - 30\gamma^2\zeta_3 + 20\zeta_2\zeta_3 - 5\zeta_5. \tag{6.2}$$

The universal constant is $C_5 = 0$, a different quantity.

For the finite differences, let b, c, d instead denote the preceding index $n-1$. Then

$$\begin{aligned} \Delta p_{5,4}(n) &= \frac{5}{n}, \\ \Delta p_{5,3}(n) &= -\frac{5}{n^2} - \frac{20b}{n}, \\ \Delta p_{5,2}(n) &= -\frac{5}{n^3} + \frac{10b}{n^2} + \frac{30b^2 - 20c}{n}, \\ \Delta r_5(n) &= \frac{-20b^3 + 60bc - 20d}{n} + \frac{10c}{n^2} + \frac{20b}{n^3}. \end{aligned} \tag{6.3}$$

Equivalently the first numerator in the last line is $5r_4(n-1)$. These identities can be checked simply by replacing (b, c, d, e) by $(b+1/n, c+1/n^2, d+1/n^3, e+1/n^4)$ in (6.1); Proposition 2.2 gives an independent binomial derivation.

The higher poles therefore contribute

$$\begin{aligned} J_5 &= \sum_{n\geq 1} \left\{ \frac{30(H_{n-1} - \gamma)^2 - 20(H_{n-1}^{(2)} + \zeta_2)}{n^2} - \frac{35}{6n^4} \right\} \\ &= 30\gamma^2\zeta_2 - 60\gamma\zeta_3 + \frac{35}{3}\zeta_4. \end{aligned} \tag{6.4}$$

For completeness, the low-weight evaluations needed in this last step are

$$Z(2,1) = \zeta_3, \quad Z(2,1,1) = \zeta_4, \quad Z(2,2) = \frac{3}{4}\zeta_4, \quad \zeta_2^2 = \frac{5}{2}\zeta_4.$$

They follow from the standard duality, sum, and stuffle identities [5, 6]. In particular,

$$\begin{aligned} \sum_{n\geq 1} \frac{(H_{n-1} - \gamma)^2}{n^2} &= \frac{11}{4}\zeta_4 - 2\gamma\zeta_3 + \gamma^2\zeta_2, \\ \sum_{n\geq 1} \frac{H_{n-1}^{(2)} + \zeta_2}{n^2} &= \frac{13}{4}\zeta_4, \end{aligned}$$

which proves every scalar term in (6.4).

Applying finite stuffle to the last line of (6.3) yields the complete spectral identity

$$\begin{aligned} D_5(s) = &-120Z(1+s,1,1,1) + 120\gamma Z(1+s,1,1) \\ &+ 60(\zeta_2 - \gamma^2)Z(1+s,1) + 20(\gamma^3 - 3\gamma\zeta_2 + \zeta_3)\zeta(1+s) \\ &+ 20Z(1+s,3) + 10Z(2+s,2) + 20Z(3+s,1) \\ &+ 10\zeta_2\zeta(2+s) - 20\gamma\zeta(3+s). \end{aligned} \tag{6.5}$$

All equalities here hold first for $\Re s > 0$, and then as meromorphic germs. In the first numerator of (6.3), the words $(2,1)$ and $(1,2)$ from H_{n-1}^3 cancel against the corresponding words in $H_{n-1}H_{n-1}^{(2)}$. The residual (3) word does not cancel. It is responsible for $Z(1+s,3)$.

Set

$$D_{p,q} = [u]Z(p+u,q), \qquad \Omega_4 = 2D_{1,3} + D_{2,2} + 2D_{3,1}. \tag{6.6}$$

For $p \geq 2$ these coefficients are ordinary outer-order derivatives, $D_{p,q} = \partial_1 Z(p,q)$. For $p = 1, q = 3$, coefficient extraction is required at a simple pole.

Theorem 6.1 (Complete fifth-power finite part)**.** *In the conventions* (1.3), (1.2), (5.3), *and* (6.6),

$$\begin{aligned} Q_5 = {} & 30\gamma^2\zeta_2 - 60\gamma\zeta_3 + \frac{35}{3}\zeta_4 - 120\chi + 120\gamma\theta + 60(\zeta_2 - \gamma^2)\eta \\ & - 20(\gamma^3 - 3\gamma\zeta_2 + \zeta_3)\gamma_1 + 20D_{1,3} + 10D_{2,2} + 20D_{3,1} \\ & + 10\zeta_2\zeta'(2) - 20\gamma\zeta'(3). \end{aligned} \tag{6.7}$$

Equivalently the three mixed terms are $10\Omega_4$.

Proof. Use $C_5 = 0$, (6.4), and the linear coefficient of (6.5) in Theorem 4.1. In particular $[s]\zeta(1+s) = -\gamma_1$, which fixes the sign of its correction. The other coefficients are exactly those named in (5.3) and (6.6). Every step rests on a finite algebraic identity or the normalized analytic results already proved. □

This answers the fifth-power research question in [2]: the fourfold height-one coefficient enters with multiplier -120, and the mixed twofold coefficients and both ordinary zeta derivatives are retained explicitly. The result is not a reduction of χ or Ω_4 to ordinary zeta and Stieltjes constants.

6.1 An ordinary convergent formula for the same finite part

The residue polynomial and its derivative are

$$R_5(L) = -5L^4 + 60\zeta_2 L^2 - 90\zeta_4, \qquad R_5'(L) = -20L^3 + 120\zeta_2 L.$$

Consequently Corollary 4.4 gives the exact alternative

$$\begin{aligned} Q_5 = {} & 30\gamma^2\zeta_2 - 60\gamma\zeta_3 + \frac{35}{3}\zeta_4 + 20\gamma_4 - 120\zeta_2\gamma_2 \\ & - \sum_{n\geq 1}\left\{\Delta r_5(n) + \frac{20\log^3 n - 120\zeta_2\log n}{n}\right\}\log n. \end{aligned} \tag{6.8}$$

Here Δr_5 is the explicit harmonic expression (6.3), with b, c, d at $n-1$. It can equally be written in polygamma values at positive integers:

$$b = \psi(n), \qquad c = 2\zeta_2 - \psi'(n), \qquad d = \tfrac{1}{2}\psi''(n). \tag{6.9}$$

The final series in (6.8) converges absolutely; its summand is $O((1+\log n)^4/n^2)$. This gives a definition-free numerical route to Q_5 using only ordinary Stieltjes constants and ordinary polygamma evaluations, with no multivariate analytic continuation routine.

6.2 Explicit kernels for the height-one coefficients

The retained height-one coefficient has a concrete normalized integral representation. We give it at every depth so that the fourfold coefficient is specified independently of the digamma moment. Height-one Laurent constants and related normalized integrals have antecedents in Young's work [8, Section 4]. The proposition below records a direct Mellin subtraction with the coefficient and Stieltjes conventions fixed explicitly for this article. Let

$$\frac{1}{\Gamma(1+u)} = \sum_{j\geq 0} a_j u^j, \qquad K_d(t) = \frac{[-\log(1-e^{-t})]^{d-1}}{(d-1)!(e^t-1)}, \quad d \geq 1. \tag{6.10}$$

Proposition 6.2 (Normalized height-one Mellin coefficient)**.** *For $d \geq 1$,*

$$Z(1+u, 1^{d-1}) = \sum_{j=0}^{d} a_j u^{j-d} + L_d u + O(u^2), \tag{6.11}$$

where

$$L_d = a_{d+1} + \int_0^1 \log t \left\{ K_d(t) - \frac{(-\log t)^{d-1}}{(d-1)!t} \right\} \mathrm{d}t + \int_1^\infty \log t \, K_d(t) \, \mathrm{d}t. \tag{6.12}$$

Both integrals are absolutely convergent. In particular $L_2 = \eta$, $L_3 = \theta$, and $L_4 = \chi$.

Proof. The generating series of $H_{n-1}(1^{d-1})$ gives, for $\Re u > 0$,

$$\Gamma(1+u) Z(1+u, 1^{d-1}) = \int_0^\infty t^u K_d(t) \, \mathrm{d}t.$$

For example it follows by dividing $\mathrm{Li}_{1^{d-1}}(z) = (-\log(1-z))^{d-1}/(d-1)!$ by $1-z$, multiplying by z, and setting $z = e^{-t}$; for $d = 1$ the kernel is simply $1/(e^t - 1)$. Subtract the leading kernel on $(0, 1)$ to write this integral as $u^{-d} + B_d(u)$, with B_d holomorphic near zero. The subtraction is integrable with any fixed power of $\log t$, because its remainder is at worst a polynomial in $|\log t|$ near zero; the kernel decays exponentially at infinity.

The derivative identity

$$K_d(t) = -\frac{1}{d!} \frac{\mathrm{d}}{\mathrm{d}t} [-\log(1-e^{-t})]^d$$

shows that $B_d(0) = 0$: the integrals from ϵ to infinity and of the subtracted kernel from ϵ to 1 have difference

$$\frac{[-\log(1-e^{-\epsilon})]^d - (-\log \epsilon)^d}{d!} \longrightarrow 0.$$

Multiplication by $1/\Gamma(1+u)$ now proves (6.11), with $L_d = a_{d+1} + B_d'(0)$ exactly as in (6.12). □

For the fifth moment the coefficient used in this kernel is explicitly

$$a_5 = \frac{\gamma^5 - 10\gamma^3\zeta_2 + 20\gamma^2\zeta_3 + 15\gamma\zeta_2^2 - 30\gamma\zeta_4 - 20\zeta_2\zeta_3 + 24\zeta_5}{120}. \tag{6.13}$$

This follows by exponentiating $\log(1/\Gamma(1+u)) = \gamma u + \sum_{r\geq 2} (-1)^{r+1} \zeta(r) u^r / r$. The kernel is an exact analytic representation, not by itself a reduction of the arithmetic content of χ.

6.3 The mixed derivative and its polylogarithmic representations

The stuffle identity in a right half-plane gives

$$Z(1+u,3) = \zeta(1+u)\zeta_3 - Z(3,1+u) - \zeta(4+u).$$

Consequently

$$D_{1,3} = -\gamma_1\zeta_3 - \partial_2 Z(3,1) - \zeta'(4). \tag{6.14}$$

The inner-order derivative has an absolutely convergent defining series. In particular this transformation replaces a singular outer-order coefficient by a convergent inner-order derivative; it does not remove the nonsymmetric spectral information.

Define the Tornheim function

$$T(a,b,c) = \sum_{m,n\geq 1} \frac{1}{m^a n^b (m+n)^c}$$

in its absolute-convergence domain. A finite partial fraction calculation gives, for u near zero,

$$T(2,2,u) = 2Z(2+u,2) + 4Z(3+u,1). \tag{6.15}$$

Indeed, for $l = m+n$, summation over $1 \leq m < l$ uses

$$\frac{1}{m^2(l-m)^2} = \frac{1}{l^2}\left(\frac{1}{m^2} + \frac{1}{(l-m)^2} + \frac{2}{m(l-m)}\right)$$

and gives $2H^{(2)}_{l-1}/l^2 + 4H_{l-1}/l^3$. Thus

$$\Omega_4 = 2D_{1,3} + \tfrac{1}{2}\partial_3 T(2,2,0) \tag{6.16}$$

$$= 2(\partial_1 - \partial_2)Z(3,1) + \partial_1 Z(2,2) - 2\gamma_1\zeta_3 - 2\zeta'(4). \tag{6.17}$$

Spectral derivatives of Tornheim sums have substantial prior literature, including [9]. These equalities are useful coordinate changes; no arithmetic independence is asserted for the displayed coordinates.

There are also completely convergent one-dimensional representations:

$$D_{1,3} = -\gamma_1\zeta_3 + \sum_{n\geq 1} \frac{\zeta(3,n)\log n}{n}, \tag{6.18}$$

$$\partial_3 T(2,2,0) = \int_0^\infty \frac{\mathrm{Li}_2(e^{-t})^2 - e^{-t}\zeta_2^2}{t}\,\mathrm{d}t. \tag{6.19}$$

For (6.18), use $H^{(3)}_{n-1} = \zeta_3 - \zeta(3,n)$ and extract the linear coefficient. For (6.19), insert the Frullani identity $-\log(m+n) = \int_0^\infty (e^{-(m+n)t} - e^{-t})\,\mathrm{d}t/t$ and sum. Absolute integrability follows either from $\sum_{m,n\geq 1} \log(m+n)/(m^2n^2) < \infty$, or directly from the $O(\log t)$ integrand at zero and exponential decay at infinity. Equations (6.16), (6.18), and (6.19) connect the first mixed correction in a digamma power to a dilogarithm integral and a Hurwitz-zeta tail sum.

6.4 What scalar polynomial cancellation proves

Corollary 6.3 (A normalized fifth-degree cancellation)**.** *Use the fixed-coordinate finite parts* $Q_k = \mathrm{FP}\int_0^1 \psi(x)^k\,\mathrm{d}x$*, and put*

$$\Omega_4 = 2D_{1,3} + D_{2,2} + 2D_{3,1}, \qquad D_{p,q} = [s]\zeta(p+s,q).$$

Here $[s]$ denotes the coefficient of s in the Laurent expansion, including when a pole is present. Define

$$P_5(T) = T^5 + 5\gamma T^4 + 10(\gamma^2 + \zeta(2))T^3 + 10(\gamma^3 + 3\gamma\zeta(2) + \zeta(3))T^2. \tag{6.20}$$

Then

$$\mathrm{FP}\int_0^1 P_5(\psi(x))\,\mathrm{d}x = -120\chi + 10\Omega_4 + 10\zeta(2)\zeta'(2) + \frac{260}{3}\zeta(4) - 60\gamma\zeta(4) - 20\gamma^3\zeta(2) - 20\zeta(2)\zeta(3). \tag{6.21}$$

Among monic polynomials $T^5 + a_4T^4 + a_3T^3 + a_2T^2$ with coefficients in $\mathbb{Q}[\gamma, \zeta(2), \zeta(3)]$, the displayed polynomial is the unique one whose coefficients of θ, η, γ_1 vanish in the stated formulas for Q_2, Q_3, Q_4, Q_5, when these symbols are treated as formal coordinates.

Proof. The finite-part operation in one fixed coordinate is linear. In $Q_5 + a_4Q_4 + a_3Q_3 + a_2Q_2$, cancellation of the displayed θ, η, γ_1 coefficients gives, successively,

$$\begin{aligned} 120\gamma - 24a_4 &= 0, \\ 60(\zeta(2) - \gamma^2) + 24\gamma a_4 - 6a_3 &= 0, \\ -20(\gamma^3 - 3\gamma\zeta(2) + \zeta(3)) + 12(\gamma^2 - \zeta(2))a_4 - 6\gamma a_3 + 2a_2 &= 0. \end{aligned}$$

This triangular system has exactly the coefficients in (6.20). The $\zeta'(3)$ coefficient simultaneously becomes $-20\gamma + 4a_4 = 0$, while the coefficients of χ, Ω_4, and $\zeta(2)\zeta'(2)$ remain respectively $-120, 10, 10$. Before using any even-zeta reduction, the remaining scalar is

$$\frac{35}{3}\zeta(4) + 90\gamma\zeta(4) + 30\zeta(2)^2 - 20\gamma^3\zeta(2) - 60\gamma\zeta(2)^2 - 20\zeta(2)\zeta(3).$$

Substitute $\zeta(2)^2 = \frac{5}{2}\zeta(4)$ to obtain (6.21). □

The uniqueness assertion is deliberately relative to the displayed coefficient presentation and to the normalization excluding constant and linear terms. It does not assert arithmetic independence of $\chi, \Omega_4, \theta, \eta, \gamma_1$, or the impossibility of rewriting Ω_4 in another family of constants. In particular, "cancellation of γ_1" means cancellation of its explicit coefficient while Ω_4 is retained as a coordinate; a different representation of Ω_4 may contain γ_1. Further arithmetic reduction of the mixed-depth term remains a separate question.

7 Normalized primitives and polygamma identities

The same normalization gives a primitive, not only a finite moment. On $\mathbb{C} \setminus (-\infty, 0]$ use the principal logarithm.

Theorem 7.1 (An all-power primitive with fixed normalization)**.** *For $k \geq 1$, put*

$$\mathcal{A}_k(z) = \kappa_k z + p_{k,1}(0)\,\mathrm{Log}\,z + \sum_{j=2}^{k} \frac{p_{k,j}(0)z^{1-j}}{1-j} + \sum_{n\geq 1}\Bigg\{p_{k,1}(n)\log(1+z/n) - zP_{k,n}(0) + \sum_{j=2}^{k} p_{k,j}(n)\frac{(z+n)^{1-j} - n^{1-j}}{1-j}\Bigg\}. \tag{7.1}$$

The series converges normally on compact subsets of that slit plane, and

$$\mathcal{A}_k'(z) = \psi(z)^k, \qquad \mathcal{A}_k(1) = Q_k. \tag{7.2}$$

Its local Laurent–log expansion at zero has constant coefficient zero, which fixes the otherwise arbitrary additive constant.

Proof. Each brace in (7.1) is the integral from zero to z of $P_{k,n}(w) - P_{k,n}(0)$. For large n and z in a fixed compact set its magnitude is $O_k((1+\log n)^{k-1}/n^2)$, as follows by cancellation of its linear terms or from the bound in the proof of Theorem 2.1. The normally convergent series may therefore be differentiated, and (2.5) gives the first identity. Every term in the braces vanishes at zero and the analytic part begins with $\kappa_k z$. The Laurent and logarithmic terms outside the braces also have no constant coefficient. The definition of the original-coordinate finite part now gives $\mathcal{A}_k(1) = Q_k$. □

For $k = 1$, this is the normalized branch of $\log\Gamma(z)$, whose local expansion begins $-\operatorname{Log} z - \gamma z + \cdots$. For every higher power it gives an explicitly normalized antiderivative through harmonic pole coefficients. Ordinary integrals along paths contained in the slit plane follow by endpoint subtraction of $\mathcal{A}_k$. On other pole-free intervals the same conclusion holds after choosing a consistent analytic continuation.

Corollary 7.2 (Differentiated polygamma identities)**.** *For $m \geq 1$ and z away from the nonpositive integers,*

$$\frac{\mathrm{d}^m}{\mathrm{d}z^m}\psi(z)^k = (-1)^m \sum_{n\geq 0}\sum_{j=1}^{k} \frac{(j)_m p_{k,j}(n)}{(z+n)^{j+m}}. \tag{7.3}$$

The series is normally convergent. Its left side is explicitly

$$\sum_{m_1+\cdots+m_k=m} \frac{m!}{m_1!\cdots m_k!} \prod_{i=1}^{k} \psi^{(m_i)}(z), \tag{7.4}$$

where $\psi^{(0)} = \psi$ and $(j)_m$ is the rising factorial.

Proof. Differentiate (2.5) normally on compact subsets. The constant subtractions disappear after the first derivative and each pole gives the displayed rising factorial. The denominator exponent is at least two, so (2.4) proves normal convergence directly as well. The multinomial identity is the ordinary product rule. □

For example, with $b_n = H_n - \gamma$ the case $k = 2, m = 1$ gives

$$\psi(z)\psi'(z) = -\sum_{n\geq 0}\left\{\frac{1}{(z+n)^3} - \frac{b_n}{(z+n)^2}\right\}.$$

This agrees with the classical polygamma pole series at $k = 1$ and extends it to polynomial products without an undetermined analytic term. The full formulas apply at arbitrary complex z in their specified domains, not only at integer arguments.

8 Ordered depth generators and exact cancellations

The preceding spectral derivatives retain information about the order of the summation indices. A complementary question is whether selected combinations over *all depths* admit finite evaluations. We give such evaluations while keeping one tagged exponent arbitrary.

The resulting formulas connect ordered harmonic sums to ordinary Hurwitz-zeta derivatives, generalized Stieltjes constants, and $\log\Gamma$.

The entire interpolation used here is established: its one-endpoint form is the interpolant of Ihara, Nakamura, and Yamamoto [7], and its relative formulation already appears in the incoming independent-order report. Gamma formulas for depth-generating series have substantial antecedents, including Young's work [8]. The purpose of this section is to derive a common two-marker formula, prove its convergence and continuation, and extract explicit cancellations useful for the present research program.

8.1 Relative ordered harmonic sums

For $c > 0$ and a word $w = (s_1, \dots, s_d)$, write

$$Z_c(w) = \sum_{n_1 > \cdots > n_d \geq 0} \prod_{j=1}^{d} (n_j + c)^{-s_j}, \qquad Z_c(\varnothing) = 1.$$

All initial manipulations can be made in the smaller absolute-convergence domain $\Re s_j > 1$. Let F_c be the formal word series with these coefficients and constant coefficient one. Products concatenate words: the coefficient of w in FG is $\sum_{w=uv} F(u)G(v)$. Define

$$\mathcal{H}_{a,b} = F_a^{-1} F_b, \qquad h_{a,b}(s) = \zeta(s, b) - \zeta(s, a). \tag{8.1}$$

This specifies every coefficient by a finite algebraic expression. For example,

$$\mathcal{H}_{a,b}(s) = h_{a,b}(s), \qquad \mathcal{H}_{a,b}(s, t) = Z_b(s, t) - Z_a(s, t) - \zeta(s, a) h_{a,b}(t).$$

The identity $\mathcal{H}_{a,1}(w) = \Psi(w; a - 1)$ identifies the one-endpoint coefficients with [7, Proposition 2.1]; their joint entireness in the spectral indices is [7, Corollary 4.2]. It also proves the relative assertion without a new regularization prescription, because

$$\mathcal{H}_{a,b} = \mathcal{H}_{a,1} \mathcal{H}_{b,1}^{-1}$$

and each coefficient of the inverse is a finite polynomial in the coefficients of $\mathcal{H}_{b,1}$. We always use these entire spectral continuations. When $a = b + N$, $N \in \mathbb{Z}_{\geq 0}$, cancellation of the ordered products gives

$$\mathcal{H}_{b+N,b}(s_1, \dots, s_d) = \sum_{N > n_1 > \cdots > n_d \geq 0} \prod_{j=1}^{d} (n_j + b)^{-s_j}. \tag{8.2}$$

Thus this is an interpolation of strict finite harmonic sums, with the largest index on the left. Write 1^p for a block of p copies of 1, including the empty block when $p = 0$.

8.2 A two-marker theorem

Theorem 8.1 (One tagged exponent and two depth markers)**.** *Let $a, b > 0$, and put $\rho = \min(a, b)$. The series*

$$\mathscr{G}_s^{a,b}(x, y) = \sum_{p,q \geq 0} x^p y^q \mathcal{H}_{a,b}(1^p, s, 1^q) \tag{8.3}$$

converges normally on compact subsets of $\mathbb{C}_s \times \{|x| < \rho,\ |y| < \rho\}$. Define, initially when $\Re(s + x - y) > 1$,

$$S_s(c; x, y) = \sum_{n \geq 0} (n + c)^{1-s} \frac{\Gamma(n + c + y)}{\Gamma(n + c + 1 + x)}. \tag{8.4}$$

Then

$$\boxed{\mathscr{G}_s^{a,b}(x,y)=\frac{\Gamma(a+x)\Gamma(b)}{\Gamma(a)\Gamma(b+y)}\bigl(S_s(b;x,y)-S_s(a;x,y)\bigr).} \tag{8.5}$$

The difference on the right has an entire continuation in s, joint with the marking variables in the stated bidisc. The completed formula also has a meromorphic continuation in x,y. On $y=0$ the larger domain $|x|<a$, with arbitrary $b>0$, is a domain of normal convergence of the one-marker series. All spectral derivatives may be taken term by term on these domains.

Proof. For the finite interval (8.2), choosing the index $n+b$ of the tagged exponent and then the subsets above and below it gives

$$\begin{aligned}\mathscr{G}_s^{b+N,b}(x,y)&=\sum_{n=0}^{N-1}(n+b)^{-s}\prod_{k=n+1}^{N-1}\Bigl(1+\frac{x}{k+b}\Bigr)\prod_{k=0}^{n-1}\Bigl(1+\frac{y}{k+b}\Bigr)\\&=\frac{\Gamma(b+N+x)\Gamma(b)}{\Gamma(b+N)\Gamma(b+y)}\sum_{n=0}^{N-1}(n+b)^{1-s}\frac{\Gamma(n+b+y)}{\Gamma(n+b+1+x)}.\end{aligned} \tag{8.6}$$

Removable Gamma specializations are interpreted through the finite product in the first line.

Agreement on these finite intervals alone would not justify arbitrary endpoints. For that step use triangular matrices

$$E_c(n)=\begin{pmatrix}1+x/(n+c) & (n+c)^{-s}\\ 0 & 1+y/(n+c)\end{pmatrix},\qquad P_{c,M}=E_c(M-1)\cdots E_c(0).$$

Their upper-right entry records exactly one tagged letter, with x counting the letters before it and y those after it. Consequently the upper-right entry of $P_{a,M}^{-1}P_{b,M}$ records the corresponding relative coefficients.

To pass rigorously to exponent 1, first replace every background factor $(n+c)^{-1}$ by $(n+c)^{-\tau}$, with real $\tau>1$. The infinite products then give the coefficients $\mathcal{H}_{a,b}(\tau^p,s,\tau^q)$ in an absolute-convergence region. Fix $0<\delta<\rho/2$ and $|x|,|y|<\delta$. For $\tau\geq 1$ sufficiently close to 1, the logarithms of the diagonal *quotient* products have differences $O(n^{-\tau-1})$. After division by the first diagonal entry, each off-diagonal summand is bounded by $C(n+1)^{-\Re s+2\delta}$, locally uniformly in the parameters. Indeed its product factors have logarithmic absolute value at most $(|x|+|y|)\log(n+1)+O(1)$. Thus the products and off-diagonal sums converge uniformly for $\Re s>1+2\delta$ as $\tau\downarrow 1$. Cauchy coefficient extraction commutes with this limit. Entireness of the relative coefficients therefore identifies its coefficients with $\mathcal{H}_{a,b}(1^p,s,1^q)$.

For completeness, the first and second diagonal entries of $P_{c,M}$ are $A_c(M,x)$ and $A_c(M,y)$, where

$$A_c(M,z)=\frac{\Gamma(M+c+z)\Gamma(c)}{\Gamma(M+c)\Gamma(c+z)}.$$

Its upper-right entry is

$$\frac{\Gamma(M+c+x)\Gamma(c)}{\Gamma(M+c)\Gamma(c+y)}\sum_{n=0}^{M-1}(n+c)^{1-s}\frac{\Gamma(n+c+y)}{\Gamma(n+c+1+x)}.$$

Inserting these expressions in the inverse of a triangular matrix and using $\Gamma(M+c+z)/\Gamma(M+c)\sim M^z$ proves (8.5) on the initial region just specified.

It remains to justify the claimed continuation and the convergence of its Taylor series. The Gamma-ratio expansion [5, §5.11(iii)] is locally uniform in the markers:

$$\frac{\Gamma(t+y)}{\Gamma(t+1+x)}=t^{y-x-1}\left(\sum_{j=0}^{J-1}c_j(x,y)t^{-j}+O(t^{-J})\right),\qquad c_0=1.$$

The coefficients do not depend on the endpoint; for instance, $c_1 = (y-x-1)(y+x)/2$. Let $r_J(t)$ be the remainder after multiplying this expansion by t^{1-s}. Subtraction of the same asymptotic terms at both endpoints yields

$$S_s(b;x,y) - S_s(a;x,y) = \sum_{j=0}^{J-1} c_j(x,y) h_{a,b}(s+x-y+j) + \sum_{n\geq 0} \bigl(r_J(n+b) - r_J(n+a)\bigr). \tag{8.7}$$

The remainder series is normally convergent whenever $\Re(s+x-y)+J>1$. Each $h_{a,b}$ is entire: the two simple Hurwitz-zeta poles have the same residue and cancel. Increasing J therefore continues the difference to every s, jointly holomorphically on compact subsets of the stated bidisc. The finitely many small-n terms present no singularity there.

This continuation has Taylor coefficients in x, y that are entire in s. They agree with those in (8.3) on the initial half-plane, hence agree everywhere. The Taylor series of a holomorphic function on a bidisc converges normally on its compact subsets, proving the series assertion. The same subtraction, away from the discrete Gamma-pole sets, gives meromorphic continuation in the markers. Finally, at $y=0$ the kernel $\Gamma(t)/\Gamma(t+1+x)$ is entire in x. The only possible marking poles of the completed expression then come from $\Gamma(a+x)$. This proves holomorphy and Taylor convergence for $|x|<a$. □

8.3 Integer gaps and ordinary zeta derivatives

The gamma quotient in (8.4) becomes a polynomial when the markers have a suitable integer difference. To state every such case uniformly, define $c_{m,k}(x)$ by

$$(t+x+1)_m = \sum_{k=0}^{m} c_{m,k}(x) t^k, \qquad m \in \mathbb{Z}_{\geq 0},$$

where $(v)_m = v(v+1)\cdots(v+m-1)$ and $(v)_0 = 1$.

Corollary 8.2 (Finite reduction at an integer marker gap)**.** *For every complex s,*

$$\mathscr{G}_s^{a,b}(x, x+m+1) = \frac{\Gamma(a+x)\Gamma(b)}{\Gamma(a)\Gamma(b+x+m+1)} \sum_{k=0}^{m} c_{m,k}(x) h_{a,b}(s-1-k). \tag{8.8}$$

This is an ordinary convergent depth sum when the markers belong to the convergence domain of Theorem 8.1; elsewhere it is an identity of the specified meromorphic continuations. In particular, for an integer $m \geq 1$, $a > m$, and $b > 0$,

$$\boxed{\sum_{p\geq 0} (-m)^p \mathcal{H}_{a,b}(1^p, s) = \frac{\Gamma(a-m)}{\Gamma(a)} \sum_{k=0}^{m-1} (-1)^{m-1-k} \begin{bmatrix} m \\ k+1 \end{bmatrix} h_{a,b}(s-1-k).} \tag{8.9}$$

Here $\left[\begin{smallmatrix} m \\ j \end{smallmatrix}\right]$ is an unsigned Stirling number of the first kind. The series in (8.9) converges normally locally in s.

Proof. On $y = x+m+1$ the kernel quotient is exactly $(t+x+1)_m$. Equation (8.8) follows first where the defining Dirichlet series converge, and then everywhere by (8.7). For (8.9), set $x=-m$, $y=0$ and use

$$\frac{\Gamma(t)}{\Gamma(t+1-m)} = \prod_{j=1}^{m-1} (t-j) = \sum_{k=0}^{m-1} (-1)^{m-1-k} \begin{bmatrix} m \\ k+1 \end{bmatrix} t^k.$$

The condition $a > m$ places this specialization strictly inside the one-marker Taylor disc. □

The first two reductions, including their normal-convergence conditions, are particularly simple:

$$\sum_{p\ge 0}(-1)^p\mathcal{H}_{a,b}(1^p,s)=\frac{h_{a,b}(s-1)}{a-1}, \qquad a>1, \tag{8.10}$$

$$\sum_{p\ge 0}(-2)^p\mathcal{H}_{a,b}(1^p,s)=\frac{h_{a,b}(s-2)-h_{a,b}(s-1)}{(a-2)(a-1)}, \qquad a>2. \tag{8.11}$$

All their spectral derivatives are finite combinations of ordinary Hurwitz-zeta derivatives. At a canceled pole they become Stieltjes differences. More explicitly, use the convention

$$\zeta(1+u,c)=\frac{1}{u}+\sum_{k\ge 0}\frac{(-1)^k}{k!}\gamma_k(c)u^k.$$

Differentiating (8.10) at $s=2$ gives, for $a>1$, $b>0$,

$$\boxed{\sum_{p\ge 0}(-1)^p\,\frac{\partial^k}{\partial s^k}\mathcal{H}_{a,b}(1^p,s)\Big|_{s=2}=\frac{(-1)^k\big(\gamma_k(b)-\gamma_k(a)\big)}{a-1}.} \tag{8.12}$$

For $k=0$ this is $(\psi(a)-\psi(b))/(a-1)$. At the adjacent point $s=1$, the classical relation $\zeta'(0,c)=\log\Gamma(c)-\frac{1}{2}\log(2\pi)$ [5, §25.11] instead yields

$$\boxed{\sum_{p\ge 0}(-1)^p\,\frac{\partial}{\partial s}\mathcal{H}_{a,b}(1^p,s)\Big|_{s=1}=\frac{\log\Gamma(b)-\log\Gamma(a)}{a-1}.} \tag{8.13}$$

For $a=N+1$, $b=1$, $N\ge 1$, the sum is finite and its value is $-\log(N!)/N$. These statements evaluate ordered combinations over depth. They do not remove the nonsymmetric spectral information in each separate fixed-depth coefficient.

8.4 A finite-interval formula with several tagged exponents

Corollary 8.3 (A finite staircase of markers)**.** *Let $a=b+N$, $b>0$, $N\in\mathbb{Z}_{\ge 0}$, and let $r\ge 1$ with $a>r$. For arbitrary complex $s_1,\dots,s_r$,*

$$\begin{aligned}\sum_{k_0,\dots,k_{r-1}\ge 0}\left(\prod_{j=0}^{r-1}(j-r)^{k_j}\right)\mathcal{H}_{a,b}(1^{k_0},s_1,1^{k_1},s_2,\dots,1^{k_{r-1}},s_r)\\ =\frac{\Gamma(a-r)}{\Gamma(a)}\mathcal{H}_{a,b}(s_1-1,\dots,s_r-1).\end{aligned} \tag{8.14}$$

The displayed sum is finite: all terms of depth greater than N vanish.

Proof. Introduce independent markers $x_0,\dots,x_r$ for the blocks of ones before, between, and after the tagged exponents. The finite-product calculation in (8.6), now made at each selected tagged index $t_j=n_j+b$, gives their joint generator as

$$\frac{\Gamma(a+x_0)\Gamma(b)}{\Gamma(a)\Gamma(b+x_r)}\sum_{N>n_1>\cdots>n_r\ge 0}\prod_{j=1}^{r}t_j^{1-s_j}\frac{\Gamma(t_j+x_j)}{\Gamma(t_j+1+x_{j-1})}.$$

Set $x_j=j-r$. Every quotient inside the sum becomes one, the last marker is zero, and the prefactor becomes $\Gamma(a-r)/\Gamma(a)$. The calculation may first be made with $b>r$, so that no intermediate Gamma factor is singular. Both sides are real analytic on the connected interval $b>\max(0,r-N)$, so the identity extends throughout that interval. Equivalently one may use the finite products to fill all removable intermediate specializations. □

For example, with the same finite-interval hypothesis and $a > 2$,

$$\sum_{p,q\geq 0} (-2)^p(-1)^q \mathcal{H}_{a,b}(1^p, s, 1^q, t) = \frac{\mathcal{H}_{a,b}(s-1, t-1)}{(a-1)(a-2)}.$$

At $s = t = 2$ its value is

$$\frac{(\psi(a) - \psi(b))^2 - \zeta(2, b) + \zeta(2, a)}{2(a-1)(a-2)}.$$

Corollary 8.3 is asserted for finite intervals only. An extension to independent endpoints with several unbounded depth markers would require its own convergence argument.

8.5 A polylogarithmic companion

Use the one-variable multiple polylogarithm convention

$$\mathrm{Li}_{s_1,\dots,s_d}(z) = \sum_{n_1>\cdots>n_d\geq 1} \frac{z^{n_1}}{n_1^{s_1}\cdots n_d^{s_d}}, \qquad |z| < 1.$$

This convention places the argument on the largest index, as in the ordered harmonic sums above.

Proposition 8.4 (A depth generator and finite-depth evaluations)**.** *For $|z| < 1$ and arbitrary $s, u \in \mathbb{C}$, the series*

$$\mathscr{F}_s(z; u) = \sum_{p\geq 0} u^p \,\mathrm{Li}_{1^p,s}(z)$$

converges normally on compact subsets of these variables and satisfies

$$\boxed{\mathscr{F}_s(z; u) = (1-z)^{-u} \int_0^z \frac{(1-t)^u \,\mathrm{Li}_{s-1}(t)}{t}\,\mathrm{d}t.} \tag{8.15}$$

The branch of $\log(1-z)$ *vanishes at zero. In particular,*

$$\sum_{p\geq 0}(-1)^p \,\mathrm{Li}_{1^p,s}(z) = (1-z)\big(\mathrm{Li}_s(z) + \mathrm{Li}_{1,s-1}(z)\big), \tag{8.16}$$

$$\sum_{p\geq 0}(-2)^p \,\mathrm{Li}_{1^p,s}(z) = (1-z)^2\big(\mathrm{Li}_s(z) + \mathrm{Li}_{1,s-1}(z) - \mathrm{Li}_{1,s-2}(z)\big) + z(1-z)\,\mathrm{Li}_{s-1}(z). \tag{8.17}$$

More generally every negative integer $u = -m$ admits a finite expression using ordinary polylogarithms and $\mathrm{Li}_{1,s-j}(z)$ *with rational coefficients in z. All spectral derivatives of these identities are valid on the same disc.*

Proof. For $|u| \leq R$ and s in a fixed compact set, the absolute sum of the coefficients attached to z^n is bounded by $n^C \prod_{k=1}^n (1 + R/k) = O(n^{C+R})$ for a suitable C. Indeed one may first choose the tagged index and then independently choose the larger background indices. Summation against $|z|^n$ proves normal convergence, including in u and s.

Differentiation of the defining series gives

$$\frac{\partial \mathscr{F}_s}{\partial z} = \frac{\mathrm{Li}_{s-1}(z)}{z} + \frac{u}{1-z}\mathscr{F}_s(z; u), \qquad \mathscr{F}_s(0; u) = 0.$$

Multiplication by the integrating factor $(1-z)^u$ proves (8.15). At $u = -m$ use

$$\frac{1}{t(1-t)^m} = \frac{1}{t} + \sum_{j=1}^m \frac{1}{(1-t)^j}.$$

Thus the integral is $\mathrm{Li}_s(z) + \sum_{j=1}^{m} I_j(s-1,z)$, where $I_j(v,z) = \int_0^z \mathrm{Li}_v(t)(1-t)^{-j}\,\mathrm{d}t$. Here $I_1(v,z) = \mathrm{Li}_{1,v}(z)$, and integration by parts gives the finite recursion, for $j \geq 2$,

$$I_j(v,z) = \frac{\mathrm{Li}_v(z)}{(j-1)(1-z)^{j-1}} - \frac{\mathrm{Li}_v(z) + \sum_{k=1}^{j-1} I_k(v-1,z)}{j-1}.$$

This proves the general finite-depth assertion and, at $m = 1, 2$, the two displayed formulas. Normal convergence permits all derivatives in the tagged exponent. □

These polylogarithmic formulas, like (8.9), reduce a specified depth-generating combination. They give exact primitives and spectral derivatives with controlled depth, while leaving open the separate reduction problems for individual ordered coefficients.

9 Source audit and precise corrections

9.1 Scope of the repository audit

The source baseline is ProveIt commit `b7667ae40a2957eed9a53d8559d53a3ee20dd1c1`, including the current manuscript and the incoming archives inspected for this continuation [1]. The review concentrated on the polynomially weighted resolvent, the multiple-Gamma normalization, continuous resolvent powers, collision finite parts, and the quartic power-to-harmonic bridge [2, 3]. No new false theorem was identified in the selected repository statements used here. This is a targeted mathematical audit, not a certification of every chapter of the complete manuscript. The two corrections below concern precisely identified external sources.

The finite-part convention remains the original unit coordinate: $\mathrm{FP}\int_0^1 f(x)\,\mathrm{d}x$ retains the constant coefficient of $\int_\epsilon^1 f(x)\,\mathrm{d}x$ after its explicit local power and logarithmic terms are separated. It is not an unspecified analytic regularization. Likewise, a constant coefficient in a spectral Laurent series or in a polynomial asymptotic expansion in $\log N$ is a different operation until a comparison theorem proves equality. In the present argument the normalized contour formula and finite summation by parts provide that comparison. Matching principal parts alone would leave an undetermined entire remainder and would not prove the moment identity.

9.2 A reversed pair of integrals in a Tornheim source

The author-hosted version of Bailey and Borwein's *Computation and theory of Mordell–Tornheim–Witten sums II*, dated 5 May 2014, contains a sign reversal in equation (110), on printed page 24 [9]. Its preceding equation (109) is correct. The correction is recorded specifically for this inspected dated version; no claim is made here about an independently uninspected final typesetting.

For $\Re a > 0$, put

$$A(a) = \frac{1}{\Gamma(a+1)} \int_0^1 \frac{(-\log u)^a [-\log(1-u)]}{1-u}\,\mathrm{d}u, \tag{9.1}$$

$$B(a) = \int_0^1 \mathrm{Li}_a(u) \frac{-\log(1-u)}{u}\,\mathrm{d}u. \tag{9.2}$$

The power of $-\log u > 0$ uses its real logarithm. Both integrals converge absolutely and depend holomorphically on a in this half-plane.

Proposition 9.1 (Corrected integral identity)**.** *For* $\Re a > 0$*,*

$$\boxed{\zeta(a+2) = B(a) - A(a).} \tag{9.3}$$

Equation (110) of the specified source prints $A(a) - B(a)$*.*

Proof. Write $H_m = \sum_{j=1}^{m} j^{-1}$ and $T(a,1,1) = \sum_{m,n\geq 1} m^{-a} n^{-1} (m+n)^{-1}$. For each positive integer m, the exact finite calculation

$$\sum_{n=1}^{N} \frac{1}{n(m+n)} = \frac{H_N + H_m - H_{N+m}}{m}$$

gives $\sum_{n\geq 1} 1/[n(m+n)] = H_m/m$. Therefore

$$T(a,1,1) = \sum_{m\geq 1} \frac{H_m}{m^{a+1}} = \zeta(a+1,1) + \zeta(a+2), \tag{9.4}$$

where $\zeta(s,1) = \sum_{m\geq 1} H_{m-1} m^{-s}$. These series converge absolutely when $\Re a > 0$. Expanding the polylogarithm and logarithm in (9.2) gives $B(a) = T(a,1,1)$, while

$$\frac{-\log(1-u)}{1-u} = \sum_{n\geq 1} H_n u^n, \qquad \int_0^1 u^n (-\log u)^a \, \mathrm{d}u = \frac{\Gamma(a+1)}{(n+1)^{a+1}}$$

give $A(a) = \zeta(a+1,1)$. Absolute convergence justifies all interchanges, proving (9.3). □

There is an exact elementary counterexample to the printed sign. At $a = 1$, substitution followed by the geometric series gives

$$A(1) = \int_0^1 \frac{\log v \log(1-v)}{v} \, \mathrm{d}v = \zeta(3), \qquad B(1) = \int_0^1 \frac{\log^2 v}{1-v} \, \mathrm{d}v = 2\zeta(3).$$

Thus its printed right side is $-\zeta(3)$, whereas its left side is $\zeta(3) > 0$. This corrects the integral line; it does not invalidate equation (109) or the spectral derivative identities obtained from that correct equation. The proof above also establishes the whole half-plane identity directly, without a Carlson-theorem argument.

9.3 A distinction between multiple and ordinary zeta reductions

On printed page 205 of *Nonlinear Euler sums*, Mező states that it is not known whether all the nonlinear sums under discussion can be expressed as multiple zeta values [10]. For convergent products of positive-integer generalized harmonic numbers, this target class is too broad: a finite multiple-zeta reduction follows directly from ordering their summation indices.

To make the correction precise, use the strict finite sums

$$H_n(b_1, \ldots, b_\ell) = \sum_{n \geq j_1 > \cdots > j_\ell \geq 1} j_1^{-b_1} \cdots j_\ell^{-b_\ell}.$$

For positive integers $a_1, \ldots, a_d$, expand $\prod_{i=1}^{d} H_n^{(a_i)}$ by the ordered blocks of equal indices. Every ordered set partition $(B_1, \ldots, B_\ell)$ of $\{1, \ldots, d\}$ contributes $H_n(b_1, \ldots, b_\ell)$, with $b_j = \sum_{i \in B_j} a_i$. This is the finite stuffle product. For $q \geq 2$, each resulting term satisfies

$$\sum_{n\geq 1} \frac{H_n(b_1, \ldots, b_\ell)}{n^q} = \zeta(q, b_1, \ldots, b_\ell) + \zeta(q + b_1, b_2, \ldots, b_\ell). \tag{9.5}$$

The two terms distinguish $n > j_1$ and $n = j_1$. All sums are absolutely convergent; hence this is a finite exact reduction for every such nonlinear Euler sum.

For example,

$$\sum_{n\geq 1} \frac{H_n^2}{n^2} = 2\zeta(2,1,1) + 2\zeta(3,1) + \zeta(2,2) + \zeta(4) = \frac{17}{4}\zeta(4). \tag{9.6}$$

The first equality already demonstrates the required finite multiple-zeta reduction; the last equality uses the familiar weight-four evaluations. A precise replacement for the source's wording should identify a smaller target, such as ordinary zeta values and their products, or a specified class of linear Euler sums. Formula (9.5) does not establish either stronger reduction in general. It is a classical algebraic mechanism, not a new arithmetic-independence result.

9.4 Unchanged conjecture status

The fifth-power identity and the algorithm for arbitrary integer powers establish the finite harmonic spectral formulas stated in this article. They do not prove that every retained spectral derivative reduces to ordinary constants. A nonzero coordinate in a specified formal presentation also does not establish arithmetic independence of its numerical value. The current Gaussian S_6 and revised S_8 identities in the pinned manuscript remain conjectures [1]; none of the arguments here changes their status.

10 Verification and reproducibility

The proofs above supply the analytic justification. Computation was used to check finite algebra, branch and indexing conventions, and numerical evaluations by independent routes. It was not used to replace the contour estimates, continuation arguments, or the cutoff-to-Laurent comparison.

10.1 Exact finite algebra

The accompanying `code/power_algebra.py` implements the coefficient construction and strict-word stuffle recursion with exact rational symbolic coefficients. The script `code/exact_checks.py` verifies the finite difference recurrence through power eight, and compares the harmonic-word expansion against actual finite nested sums at several distinct indices. It also checks the full fifth spectral support, the low-power scalar reductions, the closed residue polynomial, and the canonical fifth polynomial. Its output is stored in `data/exact_checks.json`; the reusable word tables are in `data/power_coefficients.json`.

These finite checks have a precise limitation. For example, verifying the recurrence at finitely many powers does not prove its assertion for arbitrary k; Proposition 2.2 does. Likewise, symbolic reduction of a finite polynomial does not justify exchanging an infinite sum with an endpoint finite part. That step is supplied by Theorems 2.1, 3.1, and 4.1.

10.2 An independent fifth-power calculation

The script `code/independent_fifth.py` imports none of the main coefficient implementation. It reconstructs the fifth pole data, the higher-pole contribution, and the cutoff constant through its own finite stuffle calculation. It then evaluates Q_5 in two ways. The first uses local Taylor subtraction of the pole of $\psi(x)^5$, so no singular numerical quadrature is performed without

subtraction. The second uses the ordinary arithmetic series (6.8), with an independently generated Bernoulli and Hurwitz-zeta tail. With 90 working decimal digits, cutoff $N = 96$, and tail powers through n^{-36}, the two routes gave

$$Q_5 \approx -8.690303962624538266295589662365653576770293544923884292 66. \tag{10.1}$$

Their absolute discrepancy was 9.60×10^{-59}. These are floating-point diagnostics, not an interval-certified enclosure. The decimal in (10.1) is therefore reported as a diagnostic approximation; the exact assertion is Theorem 6.1.

The ordered-generator checks compare the Gamma formulas with direct finite nested sums, including nonreal tagged exponents and nonintegral lower endpoints. They also check Stieltjes and log-Gamma derivative consequences, the two-tag staircase, and the polylogarithmic formulas at negative markers -1 and -2. All 23 numerical comparisons in `code/verify_ordered_generators.py` passed at 70 working digits; the largest absolute residual was 4.92×10^{-70}. Their defining finite sums and power-series evaluations are separate from the closed formulas being checked. These are again non-interval diagnostics.

10.3 How to reproduce the package

The standalone LaTeX file builds with an ordinary `pdflatex` installation and the listed standard packages; no external images, bibliography service, or shell escape is required. The modular source fragments are included to facilitate repository integration, while the single-file source is sufficient for rebuilding the PDF. Python requirements are `sympy` and `mpmath`; the versions used for the recorded checks are given in their JSON receipts. Run

```
python3 code/exact_checks.py
python3 code/independent_fifth.py
python3 code/verify_ordered_generators.py
python3 code/build_article.py
```

from the package directory. The optional Gaussian relation experiment has separate instructions and requires a C++17 compiler.

The integration directory contains a claim ledger, the pinned source revision and hashes, proposed placement notes, and the exact scope of the corrections. It deliberately distinguishes proved additions, recovered antecedents, diagnostics, and unresolved targets. Nothing in the package changes the conjecture status of unrelated source statements.

11 Further exact research questions

The strongest next questions concern finite identities and arithmetic reduction. Estimates are useful to justify operations, but are not the principal intended deliverable.

11.1 The first mixed spectral coordinate

Determine whether Ω_4 in (6.6) can be expressed in a specified algebra generated by ordinary zeta derivatives, ordinary Stieltjes constants, and explicitly named Gamma or Barnes periods. Equations (6.16)–(6.19) give three concrete starting points: nonsymmetric derivatives of double zeta functions, a Tornheim derivative, and a dilogarithm integral. An answer should identify which independent-looking coordinates are actually removed. Merely renaming the dilogarithm

integral does not constitute an arithmetic reduction; a failed integer-relation search does not prove nonreducibility.

One tractable first task is to study identities for $\partial_1 Z(2,2)$ and $(\partial_1 - \partial_2)Z(3,1)$ together, instead of trying to evaluate $D_{1,3}$ in isolation. Differentiated stuffle determines symmetric combinations, whereas the second expression is explicitly antisymmetric in the spectral directions. A new identity must address that remaining direction.

11.2 Scalar polynomials and the growth of spectral support

The normalized fifth polynomial in Corollary 6.3 cancels the displayed lower height-one coefficients, but retains Ω_4. For general k, determine the image of scalar polynomials $P(\psi)$ under the finite-word spectral map of Theorem 5.1. One can first ask a wholly finite algebraic question: in a stated presentation modulo specified stuffle and differential relations, which coordinates can be eliminated by lower powers?

A useful all-order result would give a recurrence for normalized monic polynomials and a proved description of their remaining word support. The stronger assertion that only the single highest height-one Laurent coefficient remains is not established by the degree-four cancellation and is not established here at degree five. An arithmetic simplification of the mixed coordinate could change the conclusion; formal coordinates must not be silently promoted to independent numbers.

11.3 Explicit sixth and seventh moments

The included coefficient tables through power eight make the next cases finite, but a table is not yet a well-organized theorem. Carry out Q_6 and Q_7 with a reduced, explicitly chosen spectral coordinate family. The all-power formula already supplies the boundary constants $C_6 = -\pi^6/7$ and $C_7 = 0$, and the complete Stieltjes subtraction is explicit. The remaining task is to organize the ordinary multiple-zeta values in J_k and the word derivatives in D_k without losing diagonal terms.

Compare the resulting support with the fifth mixed coordinate. Does a finite preferred family emerge at each fixed weight after all differentiated stuffle identities are imposed? The deliverable should include exact coordinate changes and reproducible rational certificates, not only a numerical basis.

11.4 A common Hurwitz shift with correct endpoint terms

For $a > 0$, seek an analogue for $\int_0^1 \psi(a+x)^k \, \mathrm{d}x$ in a common-shift multiple-Hurwitz notation. A derivation must retain both ordinary endpoints a and $a+1$. Then study the singular limit $a \downarrow 0$ using the original coordinate and compare it with Q_k. Replacing γ_j by $\gamma_j(a)$ in the unshifted formula without deriving the new pole locations and endpoint contributions does not answer this question.

The relative functions in Section 8 suggest a useful language for endpoint differences. Their spectral indices are entire, but that alone does not determine the finite part of a shifted digamma product. A precise theorem should specify the shift in every denominator and prove the connection.

11.5 Differential polynomials in digamma and polygamma

Extend the method from ψ^k to an arbitrary fixed differential polynomial $P(\psi, \psi', \dots, \psi^{(r)})$ with constant coefficients. Its local pole data are finite and its large-circle growth is controlled, suggesting a generalized normalized expansion and a finite-part reduction. Derivatives may change the angular constant substantially, so it should be computed rather than assumed to be the same C_k.

A first exact target is the family FP $\int_0^1 \psi(x)^a \psi'(x)^b \, \mathrm{d}x$ with integers $a, b \geq 0$. Integration by parts gives relations between these moments, and the pole-coefficient method supplies their boundary terms. Classifying the independent moments after these identities would connect the primitive formulas to a broader polygamma calculus.

11.6 Several tags at independent endpoints

Corollary 8.3 is proved only for finite intervals. For independent positive endpoints a, b, construct the full upper-triangular matrix generator with several tagged exponents, prove normal convergence in all depth markers, and determine whether the integer staircase still collapses to a shifted ordered relative sum. A direct matrix and asymptotic subtraction argument would be preferable to interpolation from integer intervals, which does not by itself prove the extension.

If such a theorem holds, differentiating at several spectral poles should produce exact combinations of multivariate harmonic Laurent data. The useful target is an explicitly evaluated mixed derivative combination, with its continuation independent of the order of taking removable limits.

11.7 Boundary markers and removable Gamma singularities

The one-marker theorem provides an ordinary depth sum at $x = -m$ when $a > m$. Study the boundary $a = m$ and the exceptional points $a = m - \ell$ through the actual generating function, separating ordinary convergence, meromorphic continuation, and finite part. At finite intervals the generating function is a polynomial and some apparent Gamma poles are removable; for independent endpoints this cancellation needs a proof. Determine the finite limiting identities and their log-Gamma or Stieltjes corrections in each admissible case.

11.8 Polylogarithmic cancellation at roots of unity

The identities of Proposition 8.4 hold normally on $|z| < 1$. Analyze their boundary continuation at $z = 1, -1, i, -i$, with explicit conditions on the tagged exponent. At $z = 1$ the prefactors and divergent depth sums interact, so a boundary identity must be derived with a specified limiting operation. At other roots of unity, seek exact colored multiple-zeta formulas and their spectral derivatives. A rigorous boundary theorem may supply relations useful to the Gaussian part of the manuscript.

11.9 Gaussian relations beyond the searched family

The computation in Appendix A shows only that one explicitly generated finite family does not close the current S_6 target modulo one prime. A productive next step is to derive new exact functional equations or construct the full relevant Cayley ideal at a higher filtration, then seek a rational certificate for the target. Increasing a numerical precision or repeating

the same finite-field search does not supply a proof. The revised S_8 target requires its own exact treatment and must not inherit a proved label from a fitted expression.

11.10 Formal verification of the analytic normalization

The finite coefficient recurrence and the strict-word conversion are suitable for small proof-assistant components. The most valuable analytic components would formalize the full-circle digamma estimate, normal convergence of the subtracted Mittag–Leffler series, and the cutoff-to-Laurent lemma. Together they would make the all-power theorem reusable without re-proving an endpoint regularization separately at each degree. An exact algebra certificate alone should not be advertised as formal verification of the analytic theorem.

A A bounded investigation of the current Gaussian S_6 target

For clarity, the searched target is recorded independently of the proved digamma results. In this appendix use the colored convention

$$\mathrm{Li}_{a,b}(z_1,z_2)=\sum_{n>m\geq 1}\frac{z_1^n z_2^m}{n^a m^b},\qquad g_{a,b}=\Im\,\mathrm{Li}_{a,b}(i,1),$$

and $\beta(s)=\sum_{n\geq 0}(-1)^n/(2n+1)^s$. The current short residual in the pinned manuscript is

$$\begin{aligned}R_6 &= S_6+\frac{12334}{527}g_{6,1}+\frac{384}{31}g_{5,2}+\frac{2136}{527}g_{4,3}\\ &\quad-\frac{3179}{5713920}\pi^7+\frac{801}{2108}\beta(4)\zeta(3)+2\beta(6)\log 2,\\ S_6 &= \sum_{n\geq 0}\frac{(-1)^n H_n}{(2n+1)^6}.\end{aligned}\tag{A.1}$$

The conjecture is $R_6=0$; no such equality is proved here.

We extended the explicitly generated weight-seven relation family to maximum total input depth five. After imaginary projection, primitive integer normalization, and duplicate removal, the family had the following size:

Relation family	Rows
Double-shuffle relations	14 908
Distribution relations and specified product lifts	4 938
Lifts of the proved S_2/S_4 relations	128
Balanced Cayley relations and specified lifts	2 322
Total	22 296

There were 8 786 occupied imaginary word coordinates. A completed sparse elimination over $\mathbb{F}_{2147483647}$ found rank 8 705 and left a nonzero 16-coordinate representative of the target. The family was exhausted in approximately 328 seconds on the machine used, before the stipulated 600-second cap. Timing is not a mathematical invariant; the field, row family, rank, and nonzero reduced target are the reproducible diagnostic.

The imaginary evaluation of the exact integer target is $128588R_6$. A separate rational-arithmetic script verifies this normalization, including the conversion $\beta(7)=61\pi^7/184320$. The family generator, field eliminator, complete-run receipts, normalization check, and reproduction

note are included in `exploratory/S6_depth5_bounded_search.zip`. The raw matrix is regenerated by the included code.

This establishes nonmembership in the *specified finite-field row span*. It does not establish rational-span nonmembership: an integer matrix can lose rank modulo the selected prime. It does not establish nonvanishing of R_6, and it does not prove the conjecture. Moreover, the finite balanced Cayley family here is not claimed to generate the full Cayley shuffle ideal at depth five. This distinction prevents comparison with stronger exact obstruction statements in the incoming material from being overstated. The retained log is a reproducible computation record, not a separately replayed compact rational certificate.

A.1 A useful local attribution

The canonical S_2 section would benefit from a citation immediately beside its classical formula

$$S_2 = 2\Im \operatorname{Li}_3\left(\frac{1+i}{2}\right) - G\log 2 - \frac{\pi(\log 2)^2}{16} - \frac{\pi^3}{64}, \qquad G = \beta(2). \tag{A.2}$$

Xu and Wang, Section 3.4, Example 3.13 of the inspected arXiv version 3 [11], give the identical evaluation after $R_{1,\bar{2}} = 4S_2$ is substituted. The broader manuscript already cites that work; the proposed change is a local attribution improvement, not a correction to the value or its proof.

References

[1] V. Reshetnikov and contributors, *ProveIt*, source revision `b7667ae40a2957eed9a53d8559d53a3ee20dd1c1`, inspected 11 October 2026. Manuscript and incoming reports: https://github.com/VladimirReshetnikov/ProveIt/tree/b7667ae40a2957eed9a53d8559d53a3ee20dd1c1/Analysis/Polylogarithms/docs/manuscript and https://github.com/VladimirReshetnikov/ProveIt/tree/b7667ae40a2957eed9a53d8559d53a3ee20dd1c1/docs/incoming.

[2] ProveIt research continuation, *Quartic Digamma Bridges, Harmonic Tail Moments, and Complex Resolvent Powers*, 11 October 2026. Incoming archive `ProveIt_Quartic_Tails_Resolvents_2026-10-11.zip` at the revision in [1], especially the quartic bridge and further-research Questions 1–4.

[3] ProveIt research continuation, *Barnes Resolvents and Higher Harmonic Identities*, 11 October 2026. Incoming archive `ProveIt_Barnes_Resolvents_and_Harmonic_Identities_2026-10-11.zip` at the revision in [1], especially the quartic harmonic section and its normalization.

[4] ProveIt research continuation, *Independent Orders and Exact Reductions in Polylogarithm–Zeta Calculus*, 11 October 2026. Incoming archive `ProveIt_Independent_Orders_2026-10-11.zip` at the revision in [1], especially the relative harmonic interpolation and exact Gaussian relation framework.

[5] NIST Digital Library of Mathematical Functions. Sections 5.5 (functional relations), 5.7 (series expansions), 5.11 (asymptotics and Gamma ratios), 25.11 (Hurwitz zeta), 25.12 (polylogarithms), and 25.16 (multiple zeta values). https://dlmf.nist.gov/.

[6] K. Ihara, M. Kaneko, and D. Zagier, Derivation and double shuffle relations for multiple zeta values, *Compositio Mathematica* **142** (2006), 307–338. https://doi.org/10.1112/S0010437X0500182X.

[7] K. Ihara, Y. Nakamura, and S. Yamamoto, Interpolant of truncated multiple zeta functions. arXiv:2407.20509 (2024); published in *The Ramanujan Journal* **67** (2025), Article 1. https://arxiv.org/abs/2407.20509. https://doi.org/10.1007/s11139-025-01045-2. Proposition 2.1 and Corollary 4.2 are cited using the arXiv text.

[8] P. T. Young, Series of height one multiple zeta functions, *Integers* **24** (2024), A43. https://math.colgate.edu/~integers/y43/y43.pdf. https://doi.org/10.5281/zenodo.11221606.

[9] D. H. Bailey and J. M. Borwein, *Computation and theory of Mordell–Tornheim–Witten sums II.* Author-hosted version dated 5 May 2014, inspected at https://www.davidhbailey.com/dhbpapers/mtw2.pdf. The correction in Section 9 is specifically to equation (110), printed page 24, in this version.

[10] I. Mező, Nonlinear Euler sums, *Pacific Journal of Mathematics* **272** (2014), no. 1, 201–226. https://doi.org/10.2140/pjm.2014.272.201. https://msp.org/pjm/2014/272-1/pjm-v272-n1-p09-s.pdf. The wording discussed in Section 9 is on printed page 205.

[11] C. Xu and W. Wang, *Dirichlet type extensions of Euler sums.* arXiv:2009.11704v3, 12 April 2022, Section 3.4, Example 3.13. https://arxiv.org/html/2009.11704v3.